Appendix: Declaration of AI Use

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Declaration of Generative AI Use

This assignment was completed with assistance from Claude (Anthropic) in line with Monash University's policy on the responsible and ethical use of generative AI technologies.

Scope of AI use

AI was used for the following tasks during this assignment:

- Debugging: Diagnosing and fixing a LaTeX math-mode conflict in the summary table column headers (escaping `$` signs with `\\$`).
- PDF rendering troubleshooting: Identifying that Quarto Preview serves only the last-rendered format, and resolving figure placement issues in the PDF output using `fig-pos: "H"`.
- English proof-reading: Correcting grammar, spelling, and consistency across the Introduction, Research Question, Dataset, Dataset Description, and Results sections of my written prose.

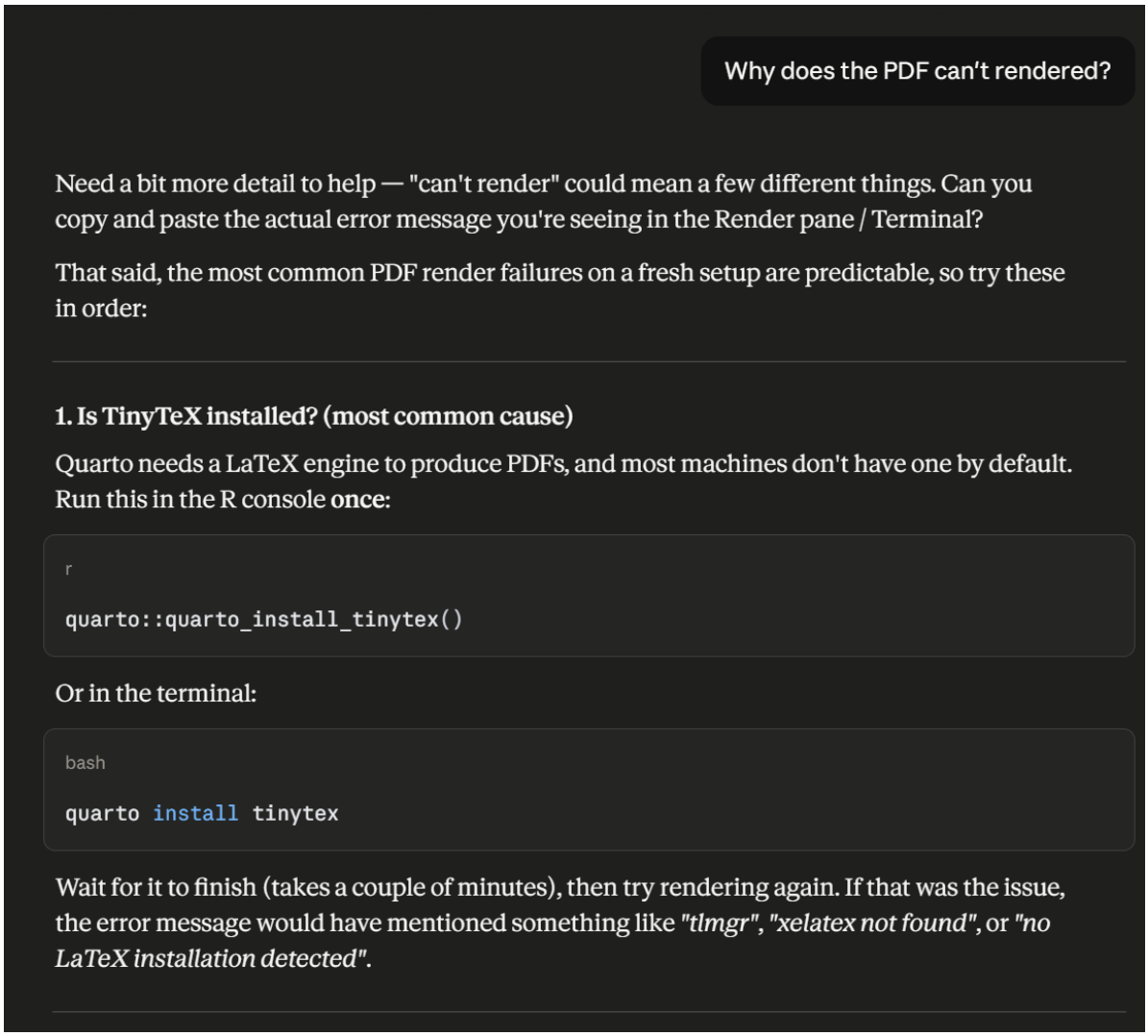
Work done independently

The following were completed by me without AI assistance:

- All prose content in the report (Introduction, Research Question, Dataset Introduction, Dataset Description commentary, Results paragraph), with AI providing only grammar and spelling corrections after I had written each section.
- The choice of research question framing, analytical approach, and interpretation of the statistical results.
- All Git and GitHub operations (clone, commit, push).

Screenshots of AI interactions

The following screenshots document my interactions with Claude relevant to this assignment.



Why does the PDF can't rendered?

Need a bit more detail to help — "can't render" could mean a few different things. Can you copy and paste the actual error message you're seeing in the Render pane / Terminal?

That said, the most common PDF render failures on a fresh setup are predictable, so try these in order:

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1. Is TinyTeX installed? (most common cause)

Quarto needs a LaTeX engine to produce PDFs, and most machines don't have one by default. Run this in the R console **once**:

```r
quarto::quarto_install_tinytex()
```

Or in the terminal:

```bash
quarto install tinytex
```

Wait for it to finish (takes a couple of minutes), then try rendering again. If that was the issue, the error message would have mentioned something like *"tlmgr"*, *"xelatex not found"*, or *"no LaTeX installation detected"*.

Figure 1: Screenshot_1:

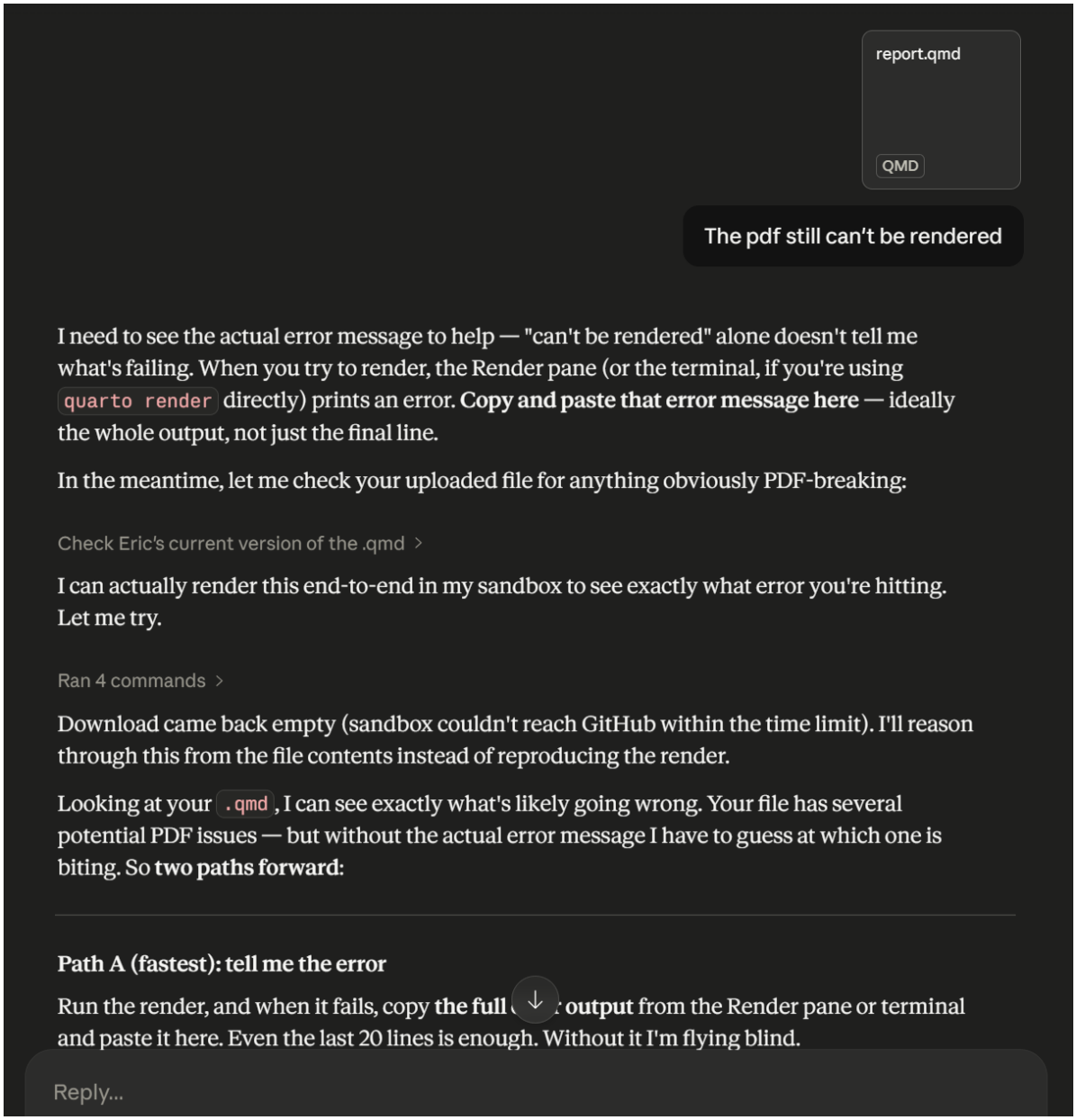


Figure 2: Screenshot_2:

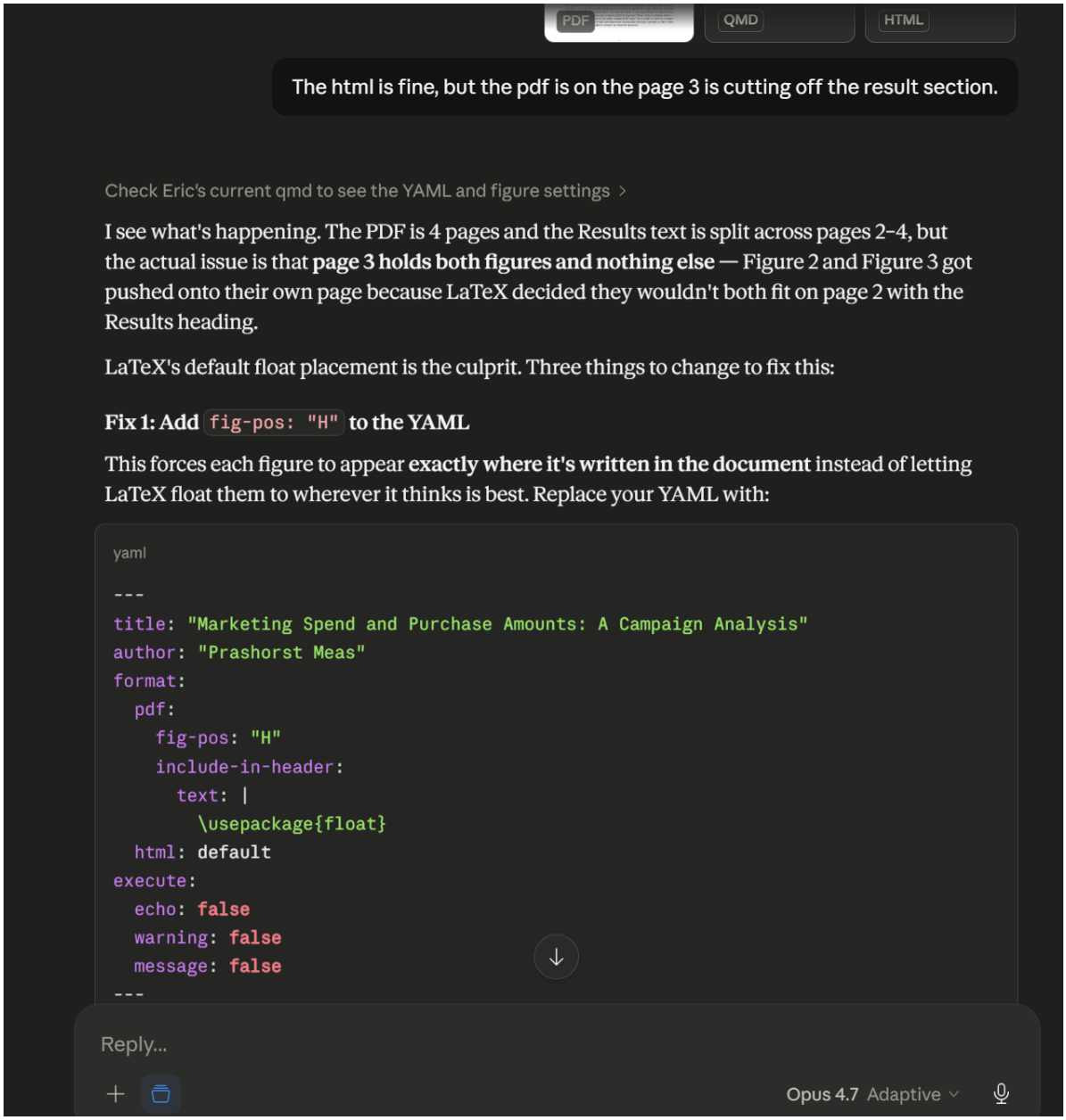
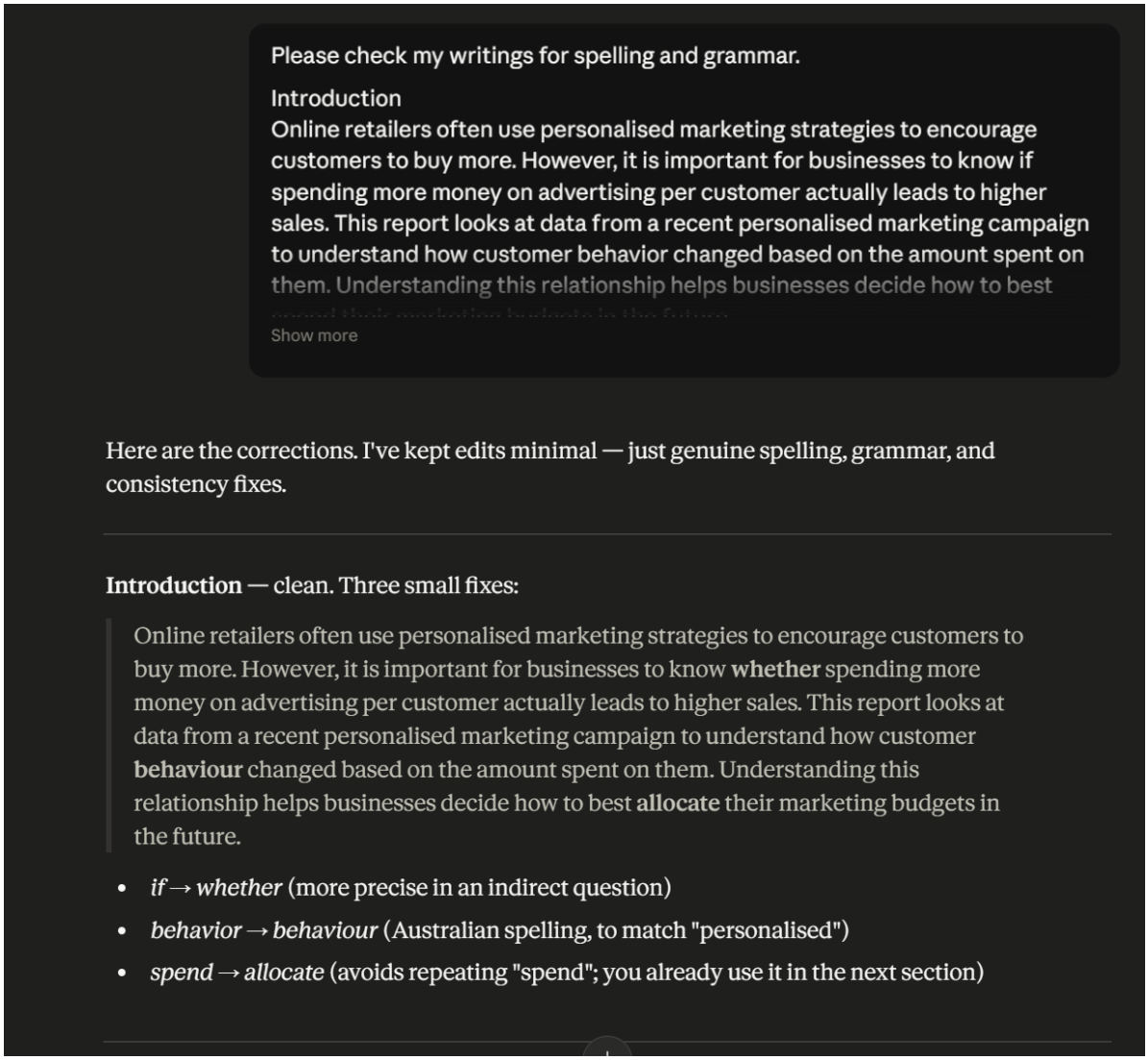


Figure 3: Screenshot_3:

Figure 4: Screenshot_4:

Figure 5: Screenshot_5:



Please check my writings for spelling and grammar.

Introduction
Online retailers often use personalised marketing strategies to encourage customers to buy more. However, it is important for businesses to know if spending more money on advertising per customer actually leads to higher sales. This report looks at data from a recent personalised marketing campaign to understand how customer behavior changed based on the amount spent on them. Understanding this relationship helps businesses decide how to best

Show more

Here are the corrections. I've kept edits minimal — just genuine spelling, grammar, and consistency fixes.

Introduction — clean. Three small fixes:

> Online retailers often use personalised marketing strategies to encourage customers to buy more. However, it is important for businesses to know **whether** spending more money on advertising per customer actually leads to higher sales. This report looks at data from a recent personalised marketing campaign to understand how customer **behaviour** changed based on the amount spent on them. Understanding this relationship helps businesses decide how to best **allocate** their marketing budgets in the future.

- *if* → *whether* (more precise in an indirect question)
- *behavior* → *behaviour* (Australian spelling, to match "personalised")
- *spend* → *allocate* (avoids repeating "spend"; you already use it in the next section)

Figure 6: Screenshot_6:

Research Question — one consistency fix:

> This report aims to answer whether a higher marketing spend per customer results in a higher purchase amount. This is an important question because it helps determine whether the **personalised** marketing strategy is profitable and should be continued. If there is a strong positive link, the business might increase marketing spend; if not, **it** might need to try cheaper advertising methods.

- *personalized → personalised* (matches the Introduction — pick one and stick with it)
- *they → it* ("business" is singular, so the pronoun should be too)
- comma before "but if not" replaced with a semicolon — smoother, and avoids a comma splice risk

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Dataset Introduction — clean prose. Two small fixes:

> The dataset records information from a recent marketing campaign, **with** one row for every customer contacted. It includes **details such as** marketing spend, site visits, purchase amounts, customer segment, and outreach notes. To clean the data, I first **standardised** the signup dates that were in different formats and treated empty strings and "N/A" text as missing values. **I then** replaced missing values in the notes column with "none" so it could be used as a proper category. Finally, I removed any rows that were missing the purchase amount, as this is the main outcome variable needed to answer my research question.

- *featuring → with* (lighter, more natural)
- *like → such as* (more formal register for a report)
- *standardized → standardised* (Australian spelling, consistent with the rest)
- *Then, I replaced → I then replaced* (reads better and avoids starting three consecutive sentences with "To clean...", "Then,...", "Finally,...")

↓

Reply...

Figure 7: Screenshot_7:

Dataset Description — clean. One small fix:

> The cleaned marketing dataset contains `` `r nrow(marketing)` `` observations across `` `r ncol(marketing)` `` variables. The variables include numeric **fields** like marketing spend and purchase amount, as well as categorical **fields** like customer segment and notes.
> As **shown** in @tbl-segment-summary, the majority of **customers** belong to the Consumer segment, while the Enterprise segment has very few customers. @fig-segment-notes shows that most customers in all segments did not have any outreach notes recorded.

- *numeric data / categorical data → numeric fields / categorical fields* (the columns are fields within the dataset; "data" is slightly awkward twice in a row)
- *seen → shown* (standard phrasing for cross-references to figures/tables)
- *the majority of the customers → the majority of customers* (drop the article — it reads cleaner)

Results — grammatically clean. Three small fixes:

> Based on the data, there is a strong positive relationship between marketing spend and how much a customer purchases, as shown in @fig-spend-purchase. Overall, the correlation is `` `r round(r_overall, 2)` ``, meaning that as marketing spend increases, the purchase amount tends to increase as well. Specifically, for every extra dollar spent on marketing, the purchase amount increases by about `` `r round(slope, 2)` `` dollars. When we look at the data split by group in @fig-spend-purchase-segment, **this positive trend is very clear for both the Consumer and SMB segments.** The Enterprise segment also shows a positive trend, but the data **are** much more spread out, likely because there are very few customers in that group. Overall, to answer the research question, higher marketing spend is **clearly** associated with higher purchase amounts, suggesting that the **personalised** marketing campaign was effective.

- *"we can see this positive trend."* → trimmed the "we can see" — the sentence is stronger

Reply...

Figure 8: Screenshot_8:

Results — grammatically clean. Three small fixes:

> Based on the data, there is a strong positive relationship between marketing spend and how much a customer purchases, as shown in @fig-spend-purchase. Overall, the correlation is `` `r round(r_overall, 2)` ``, meaning that as marketing spend increases, the purchase amount tends to increase as well. Specifically, for every extra dollar spent on marketing, the purchase amount increases by about `` `r round(slope, 2)` `` dollars. When we look at the data split by group in @fig-spend-purchase-segment, **this positive trend is very clear for both the Consumer and SMB segments.** The Enterprise segment also shows a positive trend, but the data **are** much more spread out, likely because there are very few customers in that group. Overall, to answer the research question, higher marketing spend is **clearly** associated with higher purchase amounts, suggesting that the **personalised** marketing campaign was effective.

- *"we can see this positive trend..."* → trimmed the "we can see" — the sentence is stronger without it
- *data is* → *data are* (technically "data" is plural; your unit may accept either, but "data are" is safer in a formal report)
- *definitely* → *clearly* ("definitely" reads as overconfident for a report with n=132 and an unexplained R^2 gap of ~0.34)
- *personalized* → *personalised* (consistency again)

One substantive comment (grammar-adjacent, but worth raising)

You might want to soften the final sentence of Results slightly. You're claiming the campaign "was effective," but your analysis shows a correlation between spend and purchase — not that the campaign *caused* higher purchases versus no campaign at all. A phrasing like *"suggesting that higher personalised spend is associated with larger purchases"* stays within what your data actually shows. That's a report-quality point the rubric would pick up on, not a grammar one — so take it or leave it.

Want me to check anything else?

Figure 9: Screenshot_9: